What the GTM memory experiment establishes

Pancake research companion · 22 October 2026

Pancake's Brain already represents approved GTM context. This study asks a narrower question: can selecting evidence around a Play's prerequisites help a model recommend a supported research workflow? The answer is conditional. Selection improves access to the right facts, but the model still makes enough decision errors that these results do not justify autonomous commercial execution.

The evidence

The experiment used 12 fictional company worlds, 96 queries and one shared local model. Six methods were compared under a small input allowance, followed by a two-method comparison with more room. All 768 planned comparisons are retained. They required 676 actual model requests; 92 reused a real response to an identical request. The data contain no customer records, and the workflows ran against fictional providers.

At the primary 1,024-token allowance, selecting evidence by catalog prerequisites produced **53 correct supported decisions out of 96**, compared with **40/96** for hybrid retrieval with the same domain checklist. The paired difference was 13.5 percentage points, with a 95% world-bootstrap interval from 0.0 to 28.1 points. That interval includes zero: the primary comparison does not establish clear superiority.

The more revealing distinction is between having evidence and using it. Catalog-based selection supplied complete support for every one of the 48 answerable cases. The reader produced a correctly supported recommendation in only **21/48**. It also recommended inappropriately in **16/48** cases where it should have withheld. Full context achieved 54/96 overall, while an exact SQL evaluator solved all 96 formal cases without a model.

The larger, 1,536-token comparison favored catalog-based selection: 59/96 versus 42/96, a difference of 17.7 points with an interval of 10.4–26.0. This was an exploratory comparison on the same worlds. Both methods still recommended inappropriately on 19/48 negative cases. More context did not establish reliability.

What this means for the thesis

The study supports investigating prerequisite-aware evidence selection as a component of the Brain. It does not demonstrate a new memory architecture, strategic judgment, a graph advantage or a defensible commercial moat. Generic retrieval was particularly weak on the deliberately repetitive distractors in this fixture suite; its performance here is not a verdict on every general-purpose memory system.

The experiment's useful contribution is a measurable failure boundary: a system can preserve and retrieve the right company knowledge while still choosing the wrong action. The report, data, failed attempts, source bindings and scoring code make that boundary inspectable.

Product decisions and the next evidence threshold

A concrete architecture implication

Keep approved strategy, account observations and inferred proposals distinct. Resolve explicit replacements before selection. Assemble the evidence needed for a candidate Play, then check its objective, policy versions, prerequisites and resource allowance deterministically before allowing execution. Preserve a supported proposal, a violated constraint and a request for missing evidence as different outcomes.

This division follows the observed errors. One model response cited the right records but ignored both a geography mismatch and the resource limit. Another filled the qualification requirements correctly but demanded facts needed only by unrelated alternatives. These are decision failures that more stored knowledge alone does not fix. The SQL result shows that explicit eligibility rules can be checked directly when the facts are already clean and structured.

This prototype does not measure how hard it is to extract those facts from messy documents or maintain them as a customer changes strategy. It also does not measure whether an eligible Play is worth running. Those gaps matter commercially: correct paperwork is not evidence of revenue impact.

Resources and economic interpretation

In the primary comparison, catalog-based selection used about 1,016 input tokens per query, versus 2,099 for full context. Median model-request times were 34.0 and 67.1 seconds on the tested CPU setup. Their quality difference remained uncertain; these measurements are not a guaranteed production speedup or an equal-quality cost comparison.

The two experiment blocks took 8.57 hours of elapsed server time. Local API charges were zero. Hardware value, energy, extraction, curation and human correction costs were not measured. The study therefore establishes no reduction in customer acquisition cost and no total economic advantage.

What would justify a stronger claim

The next evaluation should use independently authored histories, more varied Play alternatives and raw-document ingestion with full maintenance accounting. It should compare stronger readers and interfaces on a new frozen split, then ask blinded GTM experts whether supported proposals are useful and specific. Commercial outcomes require a separate prospective study.

For investors, the present evidence demonstrates an implemented research process and a concrete product question that can be tested. It does not establish retention, revenue lift, cross-company learning or network effects. Those remain hypotheses with separate evidence requirements.

The companion contains the English technical report, reproducible code, licensed fictional data, complete main traces and retained development failures. Its reproduction record distinguishes analysis of saved responses from a small fresh-inference check. The working repository remains private; public release and paper submission are separate milestones. This AI-assisted report has not received independent expert review or peer-reviewed acceptance.